

COSC 4806: Web Database Management

Environmental Sensor Based Modelling for Detecting and Fighting wildfire

w/ Dr. Kalpdrum Passi

Nicholas Warena / 0368871

Natural disasters are frequently occurring around the world, with several different environmental, anthropologic, and economic consequences each. Many of the most detrimental effects of climate-based disasters could be prevented or pre-emptively acted upon, but only with the proactivity that is sponsored by new technologies in the environmental sensor based modeling field. These technologies allow for the analysis and research of natural disasters, and collection of data that can help save lives. “Fire in forested areas can be regarded as an environmental disaster which is triggered by either natural forces or anthropogenic activities.” (Jaiswal et al.) For this report, wildfires will be analyzed as a major cause of loss, and a solution to prevent or eliminate small scale fires prior to their expansion will be outlined to alleviate some of the spontaneity and danger that wildfires present.

“Since September 2019, bushfires across Australia have burnt over 11 million hectares of land, killed 33 people, destroyed more than 2000 homes and displaced thousands of people. The full impact [of Australia's 2019 wildfire season] on wildlife is still to be determined.” (Wood & Ventimiglia) Unfortunately for many, wildfires are all too common in arid climates, and can happen anywhere where fuel is located. In this report, several solutions will be provided to use climate & weather data to accurately predict, and fight wildfires. With every solution comes a new set of problems, however, as the volatility of the environments in which these tragedies occur is high, solutions presented to combat loss require specific constraints, of which will be examined in detail further on in the report. Another example of a highly destructive series of wildfires are the ones that routinely ravage the Amazon rainforest of Brazil every year.

From 2013 to 2019, more than 410,000 wildfires broke out and destroyed many habitats, homes, and resulted in lower air quality, which has both short & long term impacts to human/animal health.

The main solution to be presented in this report, using environmental sensor based modelling, will attempt to create a solution to extinguish wildfires before they grow to an unmanageable level. This will be achieved by combining wireless climate detection sensors with unmanned drones to create a web of protection for areas that are more vulnerable to large-scale wildfires. With the fire prevention solution outlined in this report comes a series of new problems in order for the implementation to be successful in the average use case.

Fortunately, there has been plenty of research into the particularities of environmental sensors for modelling climatology, including simulations and implementations.

The first problem included in the wildfire prevention solution, is creating hardware that will survive, or adapt in highly volatile environments. Wireless Sensor Networks (WSNs) are currently used for monitoring/surveillance of an area, to warn of upcoming disasters, including but not limited to: wildfire[s], volcanic eruptions, flash floods, avalanches, and chemical spills. A report written in 2010 attempted to address this very issue, where the main goal of their experiment was to create a more robust net of WSNs, as many climate disasters can affect where WSNs are located, and the readings they report. "...[natural] disasters ... bear the potential to destroy the very sensor nodes that are monitoring the area to detect the disaster events. This means that sensor nodes are not available for routing of data anymore shortly after they have detected the event, e.g. they have burned in a wildfire... and therefore routes have to be changed or re-discovered to adapt to these changed conditions in order to continue the monitoring task of the network." (Wenning et al.)

A wireless sensor network consists of small, battery powered, low cost sensor nodes deployed in a forest to collect climate data such as temperature, barometric pressure, and humidity. The information is relayed to a base station, or a sink node. With this data, the base station can accurately predict and monitor wildfires, and it can do this without human-centric operations. (Aslan et al.) To properly create a network of nodes that can adapt to harsh weather conditions, the distribution of nodes must be calculated, and reinforced by clustering them into reliable grid-like patterns [Fig 2]. Having a neighbouring node allows the system to quickly implement an adaptation in the event of a node failing from the conditions, and filling each group of nodes with cluster heads allows for data aggregation and storage. And since there are differences in terrain, season and weather conditions, between every sensor, they regularly send this data to the base station for analysis.

Another problem to consider is the speed of information flow vs size of the total forest area that must be filled with sensors. “The time to detect a temperature increase at a node due to a fire is related [to] the distance of the node to the fire ignition location... to reduce the expected fire detection time, the average distance between neighboring sensor nodes should be reduced.”

(Aslan et al.) At this point, the first issue becomes more prevalent, as a dense area of nodes being destroyed would be an expensive loss, and since there are likely to be hundreds to thousands of sensors in large areas like the Amazon or Australia, recharging or replacing batteries is not a viable option, and so energy efficiency becomes the other important issue. It is more than likely that there will be limited energy storage and/or lack of energy input when deployed in remote areas, and so a solution proposed by Aslan et al is to limit the availability of energy during non-emergency hours, and to allow for uninterrupted energy use during times when it is most necessary. The simulations that the research group run reveal that nodes are more likely to use a high amount of energy in the summer, the months in which fires are most common, which is balanced by a low amount of energy in the winter.

Through their simulations, they conclude that “season, environment and weather adaptation is very important for a wireless sensor network and can significantly reduce energy consumption.” (Aslan et al.)

Since 1973, the annual amount of wildfires has slowly increased due to changes in the climate around the world, with an rising average of 70,000 per year. Adaptation within the standard fire fighting technique is inevitable, and so the last problem which will be attempted to be solved is the human labour portion of forest firefighting, by using automated drones powered with the data received from a WSN. “A forest fire usually grows exponentially and it is crucial that the fire should be detected and interfered in about six minutes to prevent the fire from spreading to a large area” (National Fire Danger Rating System (NFDRS), as cited in Aslan et al.) Getting to a fire with speed and efficiency is difficult to do as humans, however automating the process with drones can not only save time, but lives, ecosystems, and property. Unmanned Aerial Vehicle (UAV) technology could provide “an innovative forest firefighting system based on the use of a swarm of hundreds of UAVs able to generate a continuous flow of extinguishing liquid on the fire front, simulating the effect of rain.” (Ausonio et al.)

An idea proposed by Ausonio et al claims that a swarm of drones can operate without downtime, has the ability to replace exhausted batteries, and allow for multiple refills. The drones are operated by a support unit, called the “platform”, which can easily be relocated. This proposed solution provides a series of advantages, for example the ability to fight fires when common aircraft cannot, and to work flexibly and accurately in areas that are inaccessible by anthropogenic means.

Another advantage is that the amount of water used by the drones is much less than that of a firefighting aircraft, meaning that the system is extremely convenient in places where water is of low supply. The last advantage is that the proposed system uses a rain-like dispensing system, which is better suited for extinguishing fires, much like the sprinkler systems found indoors. This system is theorized to require only 50 L of water per drone to effectively fight a low-intensity, limited extent forest fire. Evidently, this is an ingenious solution.

In conclusion, combining the technologies of environmental sensors (wireless sensor networks) and a drone deployment system can result in an automated fire detection and fighting system that could save many lives, homes, and habitats, by examining and analyzing a fire while it's small, and sending a drone swarm platform to quickly send wave after wave of simulated rain to quickly eliminate the fire, even with low visibility/complicated topography. In the final report, an implementation of the grid system will be programmed, and a simulation will show how data is stored, how sensors detect fires, and how drones can attack the main area of the fire, and areas surrounding, to quickly eliminate fires with little to no human involvement. Because fires are so unpredictable, the effectiveness of drones will be an interesting research subject, and using data collected from environmental sensor based networks, solutions using this technology are readily available, to combat deadly forest fires without risking human lives. I firmly believe that with the steadily increasing cost of technology, it won't be long before this type of forest fighting is a standard, as it is more effective, more efficient, saves lives, and results in a greater quality of life for the communities supported by this technology.

Bibliography:

Aslan, YE., Korpeoglu, I., Ulusoy, O. A framework for use of wireless sensor networks in wildfire detection and monitoring, *Computers, Environment and Urban Systems*, Volume 36, Issue 6, 2012, pp. 614-625, ISSN 0198-9715,

<https://doi.org/10.1016/j.compenvurbsys.2012.03.002>

Ausonio, E.; Bagnerini, P.; Ghio, M. Drone Swarms in Fire Suppression Activities: A Conceptual Framework. *Drones* Volume 5, Issue 17, 2021

<https://doi.org/10.3390/drones5010017>

"Infographic: Record Number Of Wildfires Burning In The Amazon". Statista Infographics, 2019

<https://www.statista.com/chart/19089/number-of-wildfires-recorded-in-brazils-amazon-ra-eforest>

Jaiswal, Rajeev Kumar et al. "Forest fire risk zone mapping from satellite imagery and GIS."

International Journal of Applied Earth Observation and Geoinformation 4 (2002): 1-10

Wenning, BL., Pesch, D., Timm-Giel, A. et al. Environmental monitoring aware routing: making environmental sensor networks more robust. *Telecommun Syst* Volume 43, Issue 3. 2010

<https://doi.org/10.1007/s11235-009-9191-8>

Wood, S. Ventimiglia, A. "Global Fires". futureearth.org, (2021),

<https://futureearth.org/publications/issue-briefs-2/global-fires>